AP PRECALCULUS

Unit 1 • Worksheet 1.1-1.6C

Review 1.1-1.6C • Review of Topics 1.1 to 1.6

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1 (1.2)

NO CALCULATOR

$P(t)$ is the number of bacteria after t hours, $P(2) = 500$, $P(5) = 1400$. What is the average rate of change of P on $[2, 5]$?

(A) 300 bacteria per hour
(B) 900 bacteria per hour
(C) 280 bacteria per hour
(D) 180 bacteria per hour

Question 2 (1.4)

NO CALCULATOR

A polynomial is increasing on $(-\infty, -1)$, decreasing on $(-1, 4)$ and increasing on $(4, \infty)$. Which is true?

(A) local maximum at $x = 4$
(B) local minimum at $x = -1$
(C) local maximum at $x = -1$
(D) no local extrema

Question 3 (1.5)

NO CALCULATOR

What is the multiplicity of the zero $x = -2$ of $p(x) = (x^2 - 4)^2(x + 2)$?

(A) 1
(B) 2
(C) 3
(D) 5

Question 4 (1.6)

NO CALCULATOR

Which describes the end behavior of $p(x) = -x^2(x - 1)(x + 5)^2$?

(A) up on both ends
(B) down on both ends
(C) up on the left, down on the right
(D) down on the left, up on the right

Question 5 (Free response) CALCULATOR

The volume of a box made from a 20 cm by 30 cm sheet by cutting squares of side x cm is $V(x) = x(30 - 2x)(20 - 2x)$.

(a) What is the domain of V in this context? Explain.
(b) Find $V(3)$ and interpret it.
(c) Use a graphing calculator to find the value of x that maximizes V and the maximum volume.
(d) Find the average rate of change of V on $[1, 3]$, with units.